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(54) **REMIXING MATERIAL TRANSFER
VEHICLE WITH STORAGE CLEANOUT
DOOR**

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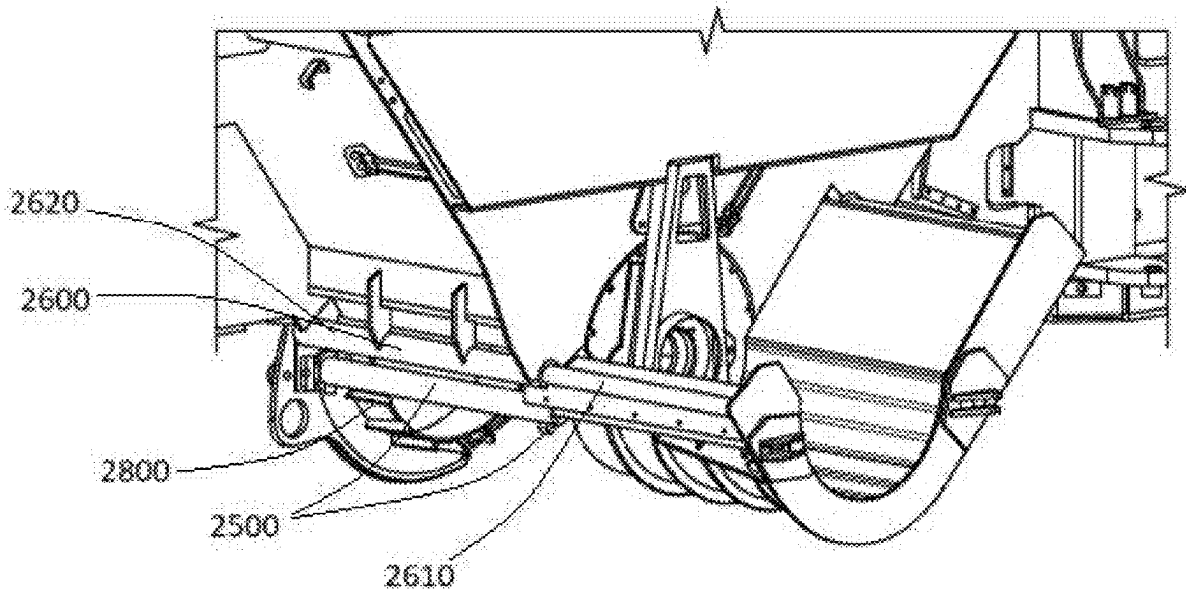
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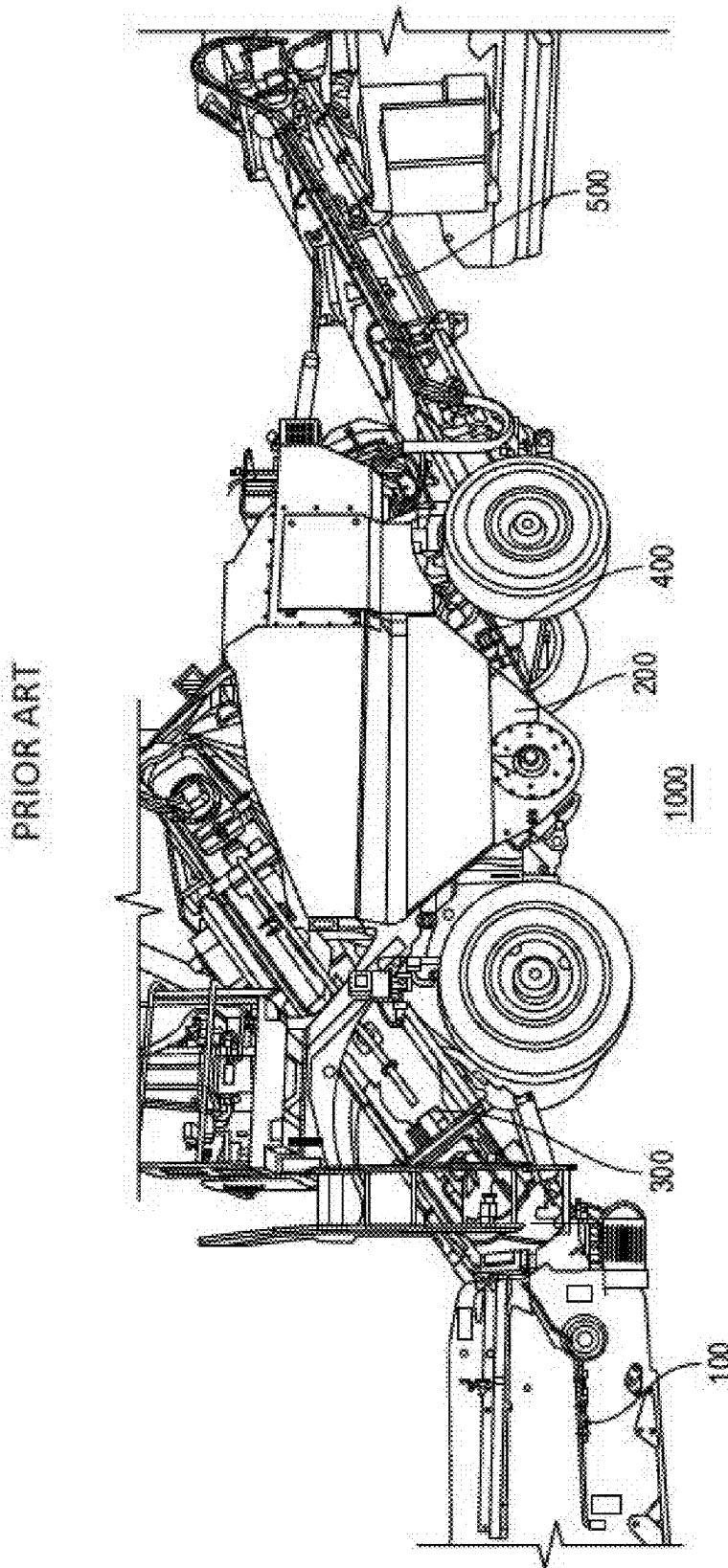
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1, 2024.

(57) **ABSTRACT**

In accordance with example embodiment, a material transfer
vehicle comprised of a hopper having a cleanout door
configured to slide out under the influence of first and second
door cylinders.





PRIOR ART

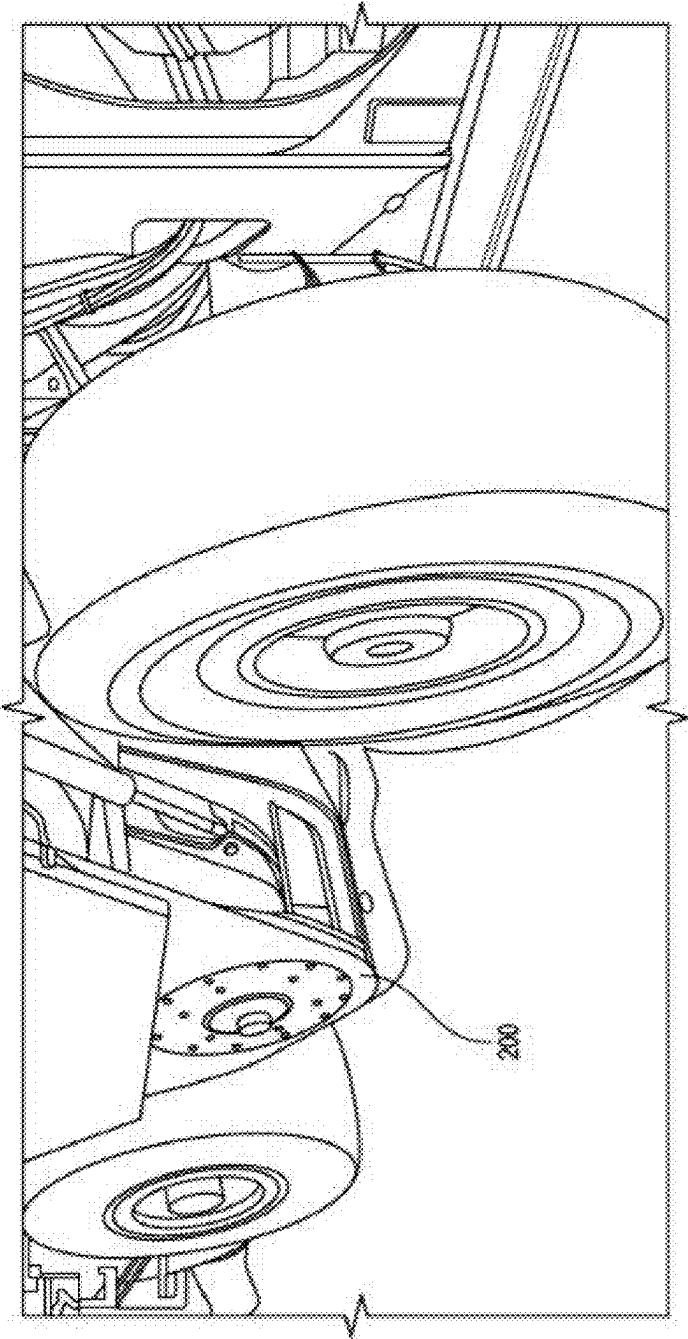
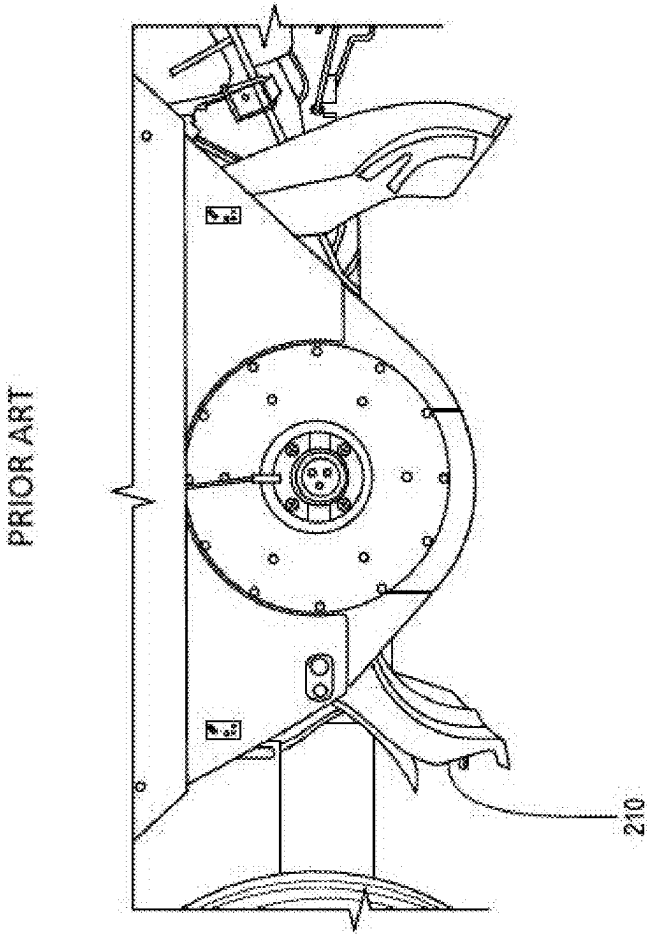


FIG. 2



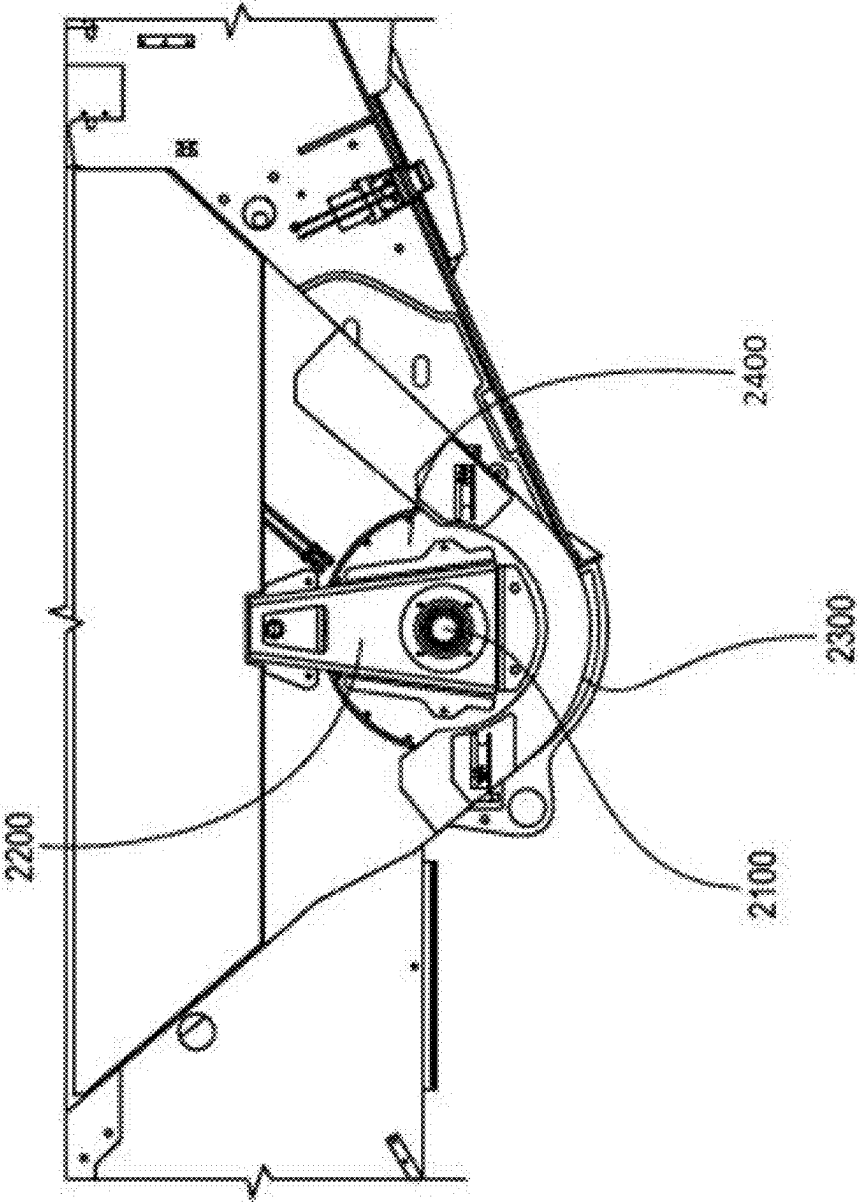
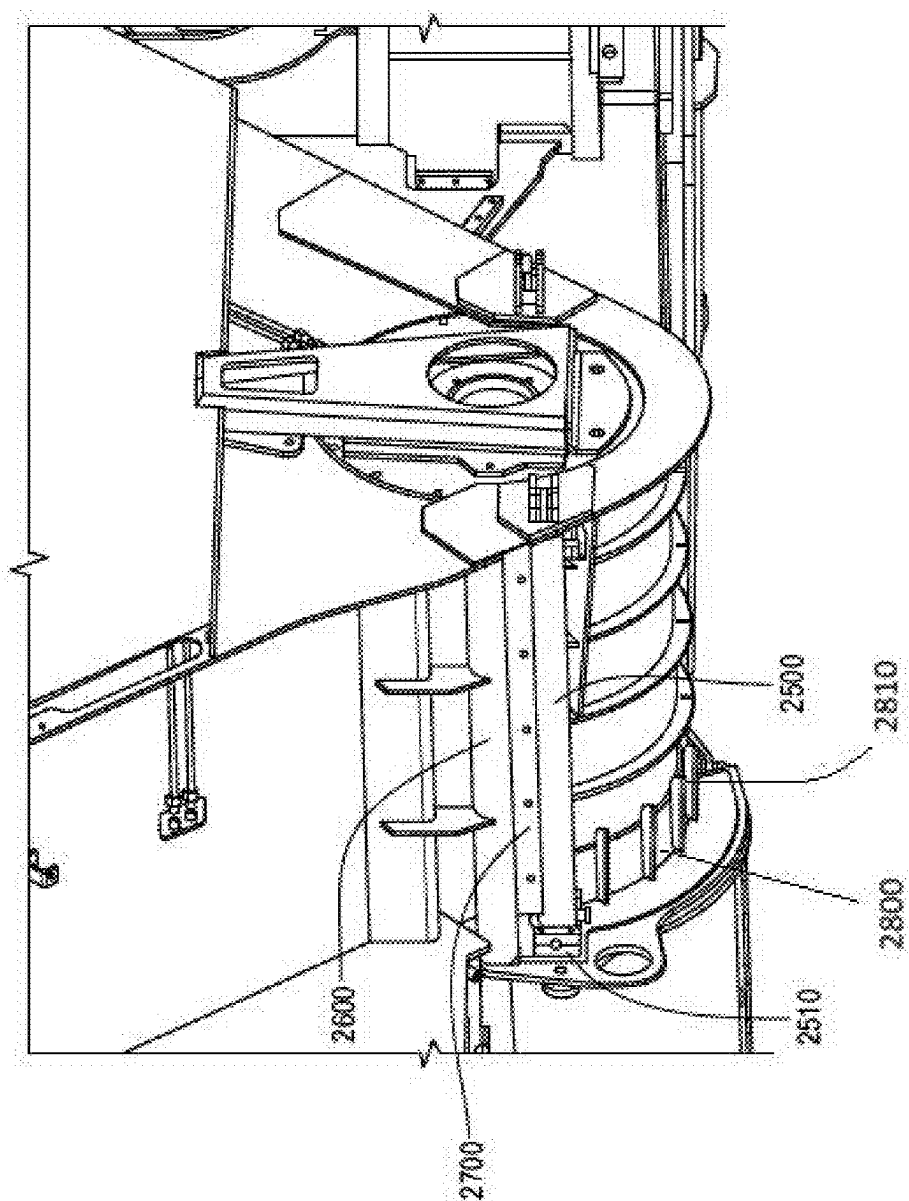


FIG. 4



501

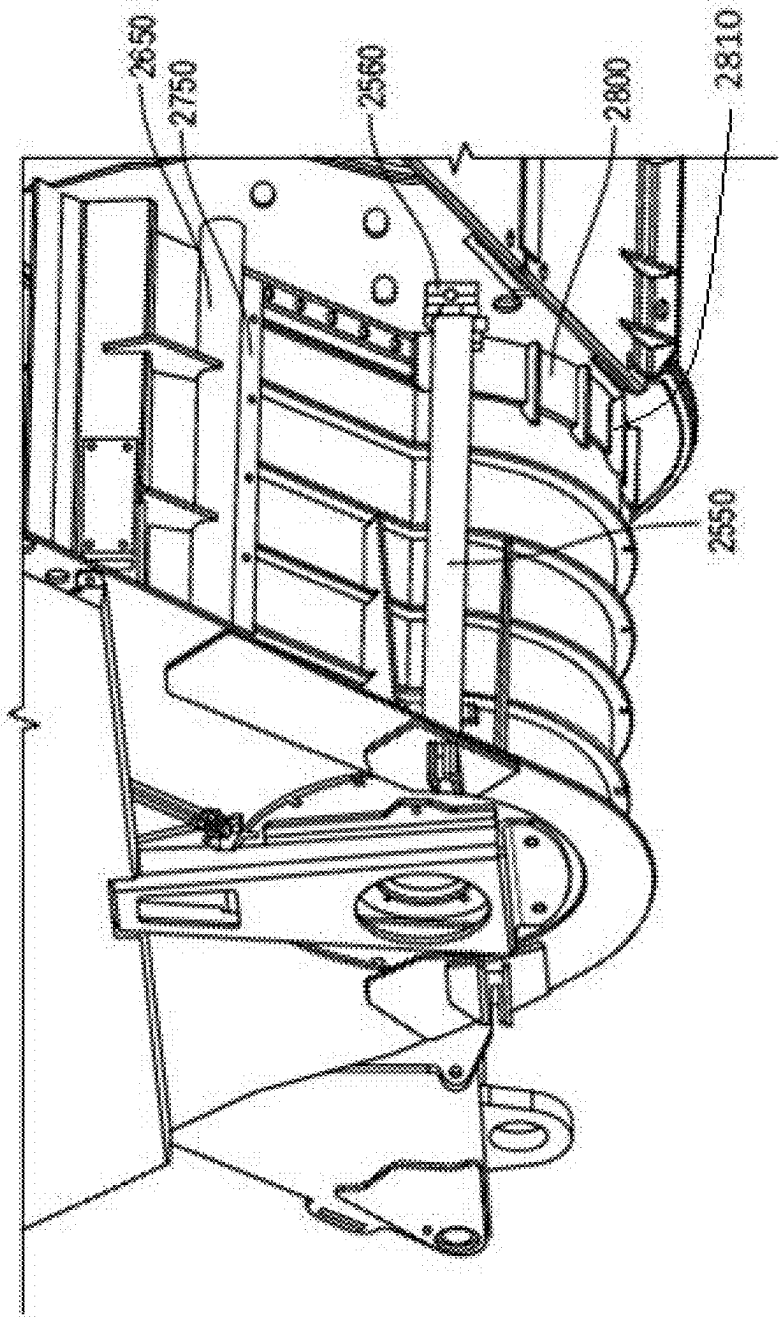


FIG. 6

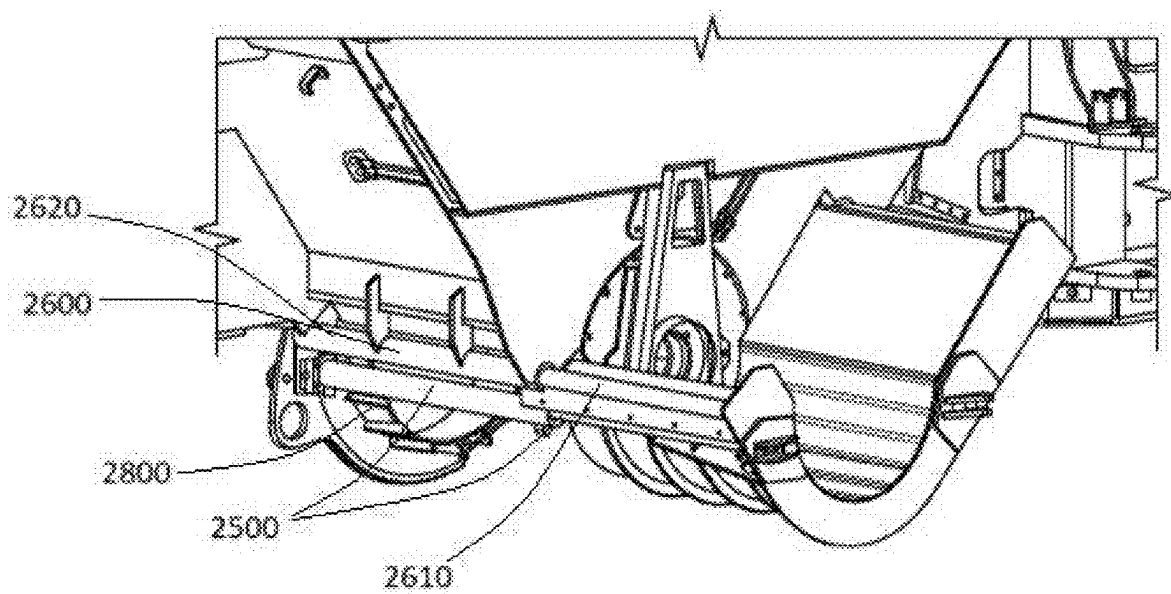


FIG. 7

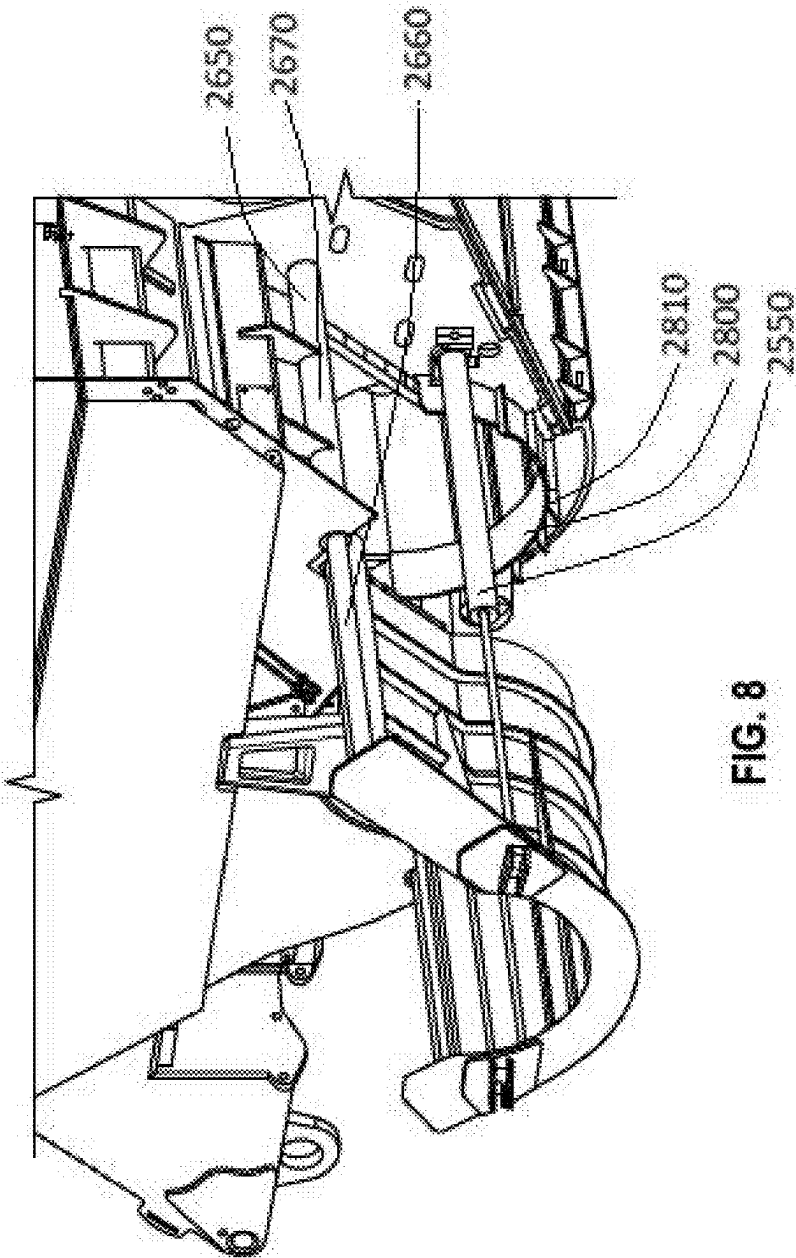


FIG. 8

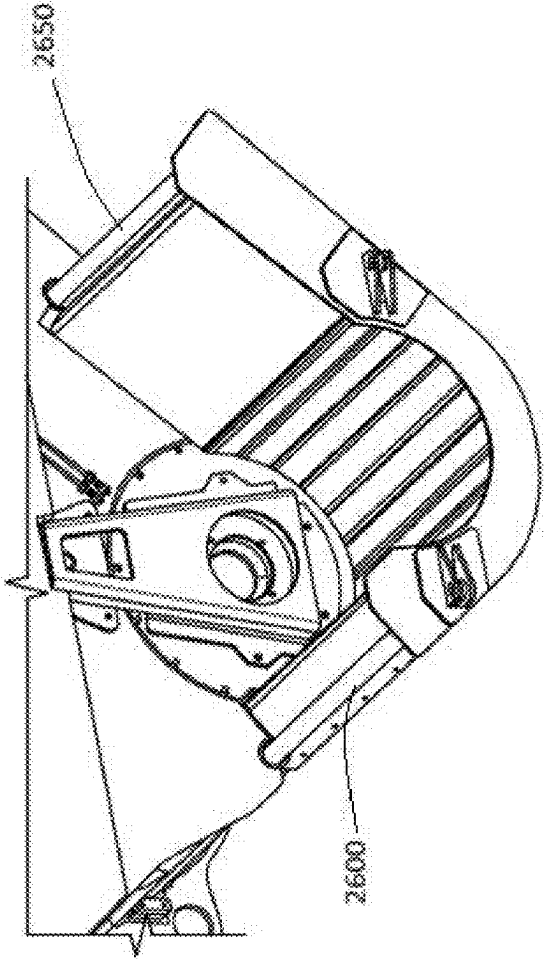
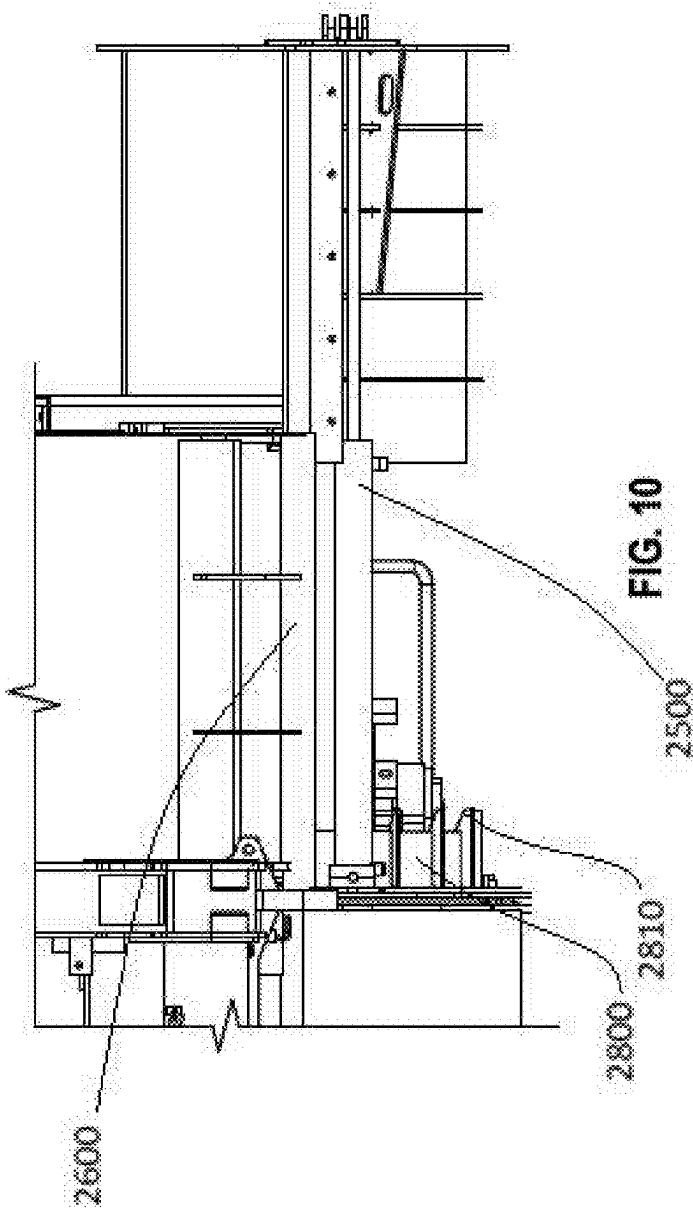


FIG. 9



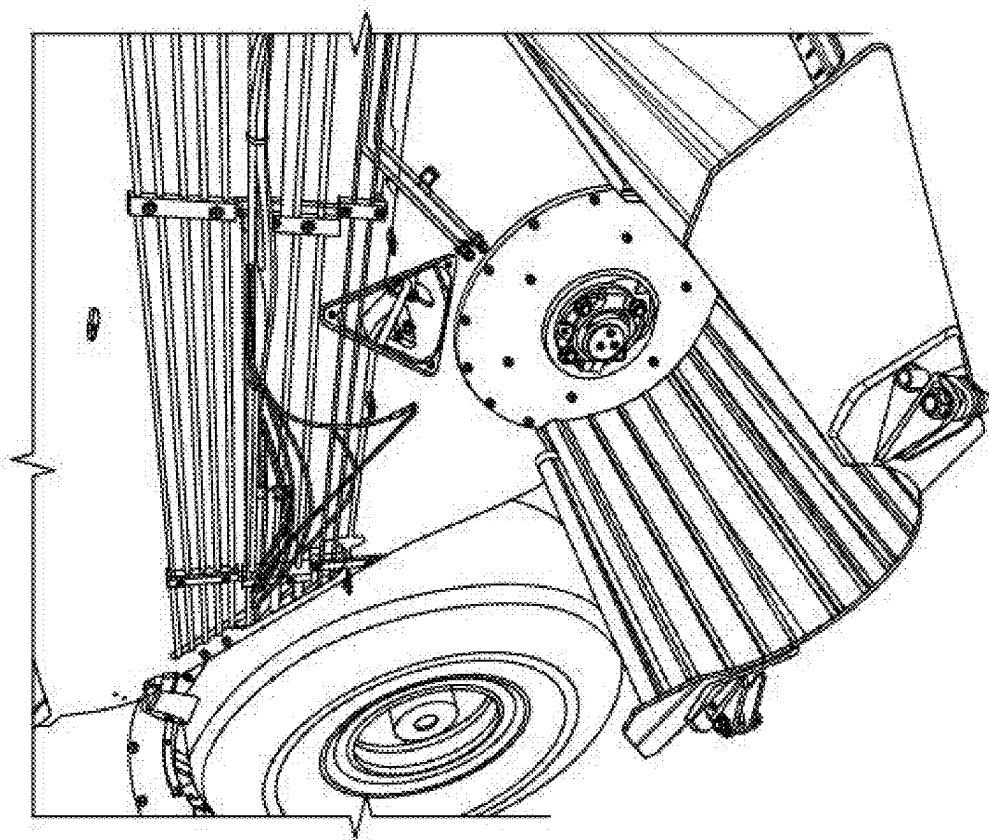


FIG. 11

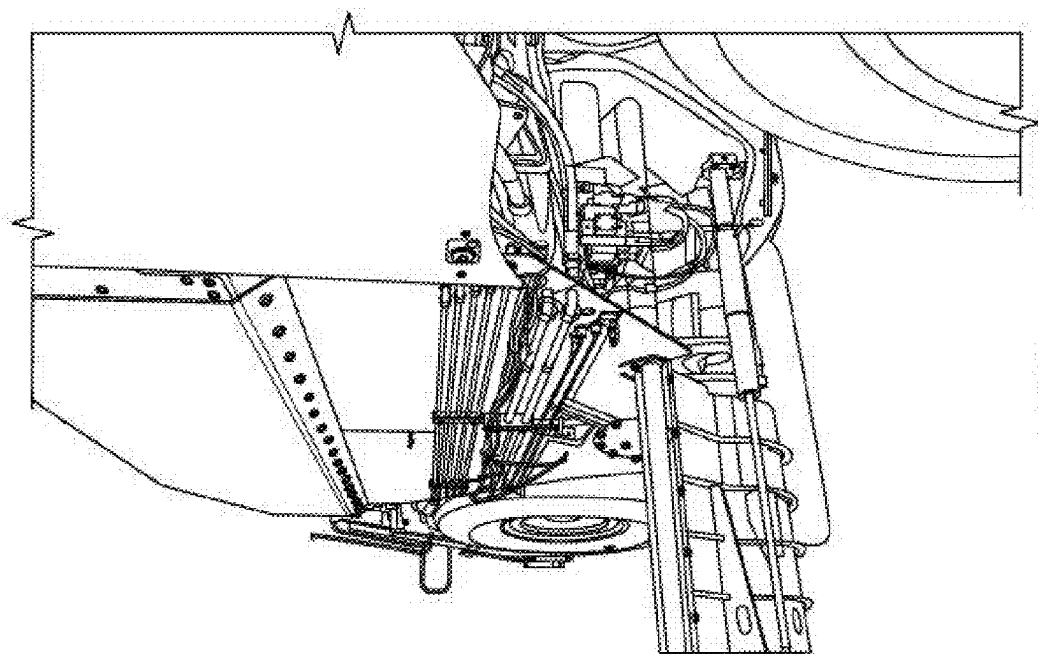


FIG. 12

FIG. 13

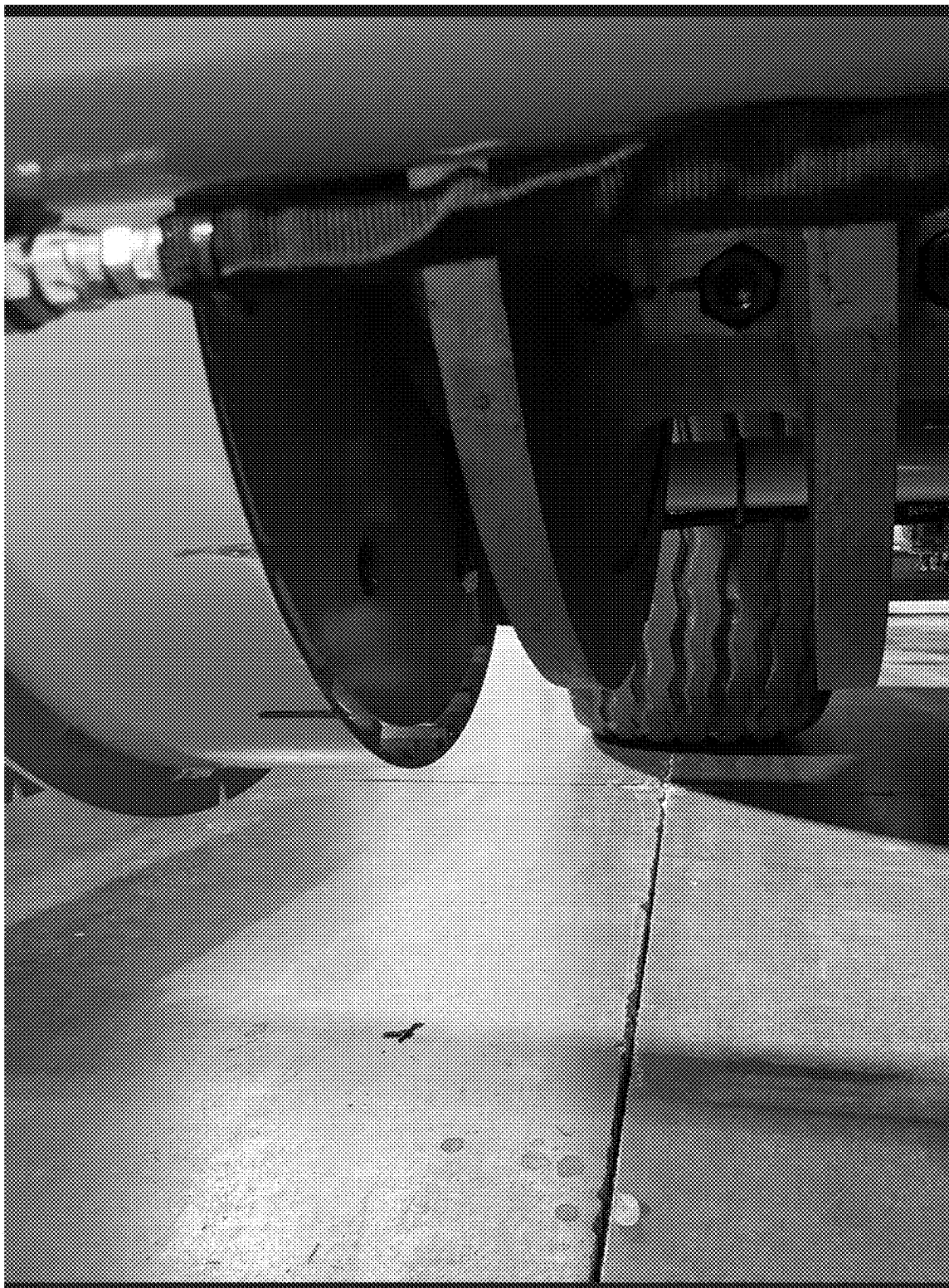


FIG. 14



REMIXING MATERIAL TRANSFER VEHICLE WITH STORAGE CLEANOUT DOOR

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/560,326 which was filed with the United States Patent and Trademark Office on Mar. 1, 2024, the entire contents of which is herein incorporated by reference.

BACKGROUND

1. Field

[0002] Example embodiments relate to a remixing material transfer vehicle having a new cleanout door.

2. Description of the Related Art

[0003] FIG. 1 is a view of a remixing transfer vehicle **1000** (an example of a material transfer vehicle). The remixing transfer vehicle **1000** generally includes a first hopper **100** into which material, for example, asphalt, may be dumped via a dump truck. The material may move out of the first hopper **100** to a second hopper **200** (generally arranged near a middle of the remixing transfer vehicle **1000**) under the influence of a first feeder **300** (which may, in one embodiment, be a chain). The second hopper **200** may temporarily store and mix/remix the material via augers that may be housed therein. The material may be moved out of the second hopper **200** by a second feeder **400** where it may flow onto a feeder chain **500** to deliver the material into another machine, for example, a paver, which may use the material to pave a road.

[0004] Generally speaking, access to the second hopper **200** may be obtained by crawling under the second hopper **200** and unlatching a latch **210**. After the latch **210** is unlatched, cylinders may be used to pivot (rotate) four clam-shell down doors downward (two of which are shown in FIG. 3). After the clam-shell doors are pivoted downwards an operator may crawl under the second hopper **200** to access its contents which may include the aforementioned augers. The augers may be serviced by the operator or the operator may remove the augers through the opening created when the clam-shell doors are pivoted downward.

SUMMARY

[0005] The inventor notes that while conventional remixing transfer vehicles like that shown in FIGS. 1-3 are extremely useful and efficient, there are some minor problems. For example, the aforementioned latch is necessary to hold the doors from creeping open. In order to open the doors the operator must crawl underneath the remixing transfer vehicle which can be quite uncomfortable. When the doors are scraped (for example, for cleaning), the operator must bend down and reach upward to clean which can be difficult to do. Additionally, when the doors open some asphalt can fall onto the ground which must be cleaned up. Additionally, when the doors open some asphalt can fall onto the ground which must be cleaned up prior to the doors being able to close. Further, asphalt can be trapped between a lower frame support and the augers making cleaning difficult. When the ground is uneven, the clamshell doors

may accidentally hit the ground when opening and closing. When the ground is uneven, the clamshell doors may be unable to open or close due to contact with the ground. After cleanout, the asphalt can harden and get in the way of the doors closing when the operator is ready to run again. Lastly, the asphalt has a chance of leaking out due to the overlapping of the doors to the frame. With the above problems in mind the inventor set out to design an improvement to a conventional remixing transfer vehicle.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Example embodiments are described in detail below with reference to the attached drawing figures, wherein:

[0007] FIG. 1 is a view of a conventional material transfer vehicle;

[0008] FIG. 2 is a close up view of a mixing hopper of the conventional material transfer vehicle;

[0009] FIG. 3 is a side view of the mixing hopper of the conventional material transfer vehicle with clam shell doors opened downwards to expose the mixing hopper;

[0010] FIG. 4 is a side view of a mixing hopper of a material transfer vehicle in accordance with an example embodiment;

[0011] FIG. 5 is a perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment;

[0012] FIG. 6 is another perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment;

[0013] FIG. 7 is a perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where a storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper;

[0014] FIG. 8 is another perspective view of the mixing hopper of the material transfer vehicle having in accordance with an example embodiment where the storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper;

[0015] FIG. 9 is another perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where the storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper;

[0016] FIG. 10 is side view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where the storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper;

[0017] FIG. 11 is another perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where the storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper;

[0018] FIG. 12 is another perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where the storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper;

[0019] FIG. 13 is another perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where the storage cleanout door of the

mixing hopper has been moved outwards to expose the mixing hopper and an auger in the mixing hopper; and

[0020] FIG. 14 is another perspective view of the mixing hopper of the material transfer vehicle in accordance with an example embodiment where the storage cleanout door of the mixing hopper has been moved outwards to expose the mixing hopper and an auger in the mixing hopper.

DETAILED DESCRIPTION

[0021] Example embodiments will now be described more fully with reference to the accompanying drawings, in which example embodiments of the invention are shown. The invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. In the drawings, the sizes of components may be exaggerated for clarity.

[0022] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it can be directly on, connected to, or coupled to the other element or layer or intervening elements or layers that may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0023] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, and/or section from another elements, component, region, layer, and/or section. Thus, a first element component region, layer or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0024] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the structure in use or operation in addition to the orientation depicted in the figures. For example, if the structure in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The structure may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0025] Embodiments described herein will refer to plan views and/or cross-sectional views by way of ideal schematic views. Accordingly, the views may be modified depending on manufacturing technologies and/or tolerances. Therefore, example embodiments are not limited to those shown in the views, but include modifications in configurations formed on the basis of manufacturing process. Therefore, regions exemplified in the figures have schematic

properties and shapes of regions shown in the figures exemplify specific shapes or regions of elements, and do not limit example embodiments.

[0026] The subject matter of example embodiments, as disclosed herein, is described with specificity to meet statutory requirements. However, the description itself is not intended to limit the scope of this patent. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different features or combinations of features similar to the ones described in this document, in conjunction with other technologies. Generally, example embodiments relate to an improved remixing transfer vehicle.

[0027] FIG. 4 is a view of a bottom of a remixing transfer vehicle 2000 in accordance with example embodiments. The remixing vehicle 2000 may be substantially identical to the remixing vehicle 1000 in that it may have a first storage hopper to receive asphalt and a chain that transfers the asphalt to a second hopper. However, as will be understood, the second hopper of the remixing transfer vehicle 2000 includes a storage cleanout door 2300 arranged at the bottom of the second hopper rather than a pair of clamshell doors as is found the remixing transfer vehicle 1000 of the conventional art.

[0028] FIG. 5 is a first perspective view of the bottom of the remixing transfer vehicle 2000 and FIG. 6 is a second perspective view of the bottom of the remixing transfer vehicle 2000. Referring to FIGS. 4-6 it is observed the remixing transfer vehicle 2000 includes a storage footshaft bearing 2100 for supporting an auger which may be located inside the remixing transfer vehicle 2000. The remixing transfer vehicle 2000 may further include a bearing guard and stiffener 2200, a bearing mount weldment 2400, and the storage cleanout door 2300. In example embodiments, the storage cleanout door 2300 may be part of a second hopper (and arranged at a bottom of the second hopper) which may receive asphalt and thus may be exposed to asphalt being remixed in the second hopper. As will be evident from the drawings and the description provided below, the storage cleanout door 2300 may slide outwards from a body of the remixing transfer vehicle 2000 thus exposing an inner surface of the storage cleanout door. This is substantially different from the clamshell doors of the conventional art 1000 which pivots (rotates) downwards.

[0029] In example embodiments the storage cleanout door 2300 may be connected to at least one actuator which may move the storage cleanout door 2300. In the exemplary embodiment of at least FIGS. 4-6 the at least one actuator may be embodied as a front storage door cylinder 2500 and a rear storage door cylinder 2510. The front storage door cylinder 2500 may have one end connected to a front cylinder trunnion mount 2510 and another end connected to the storage cleanout door 2300. Similarly, the rear storage door cylinder 2550 may have one end connected to a rear cylinder trunnion mount 2560 and another end connected to the storage cleanout door 2300. In example embodiments, the front storage door cylinder 2500 and the rear storage door cylinder 2550 may be conventional hydraulic cylinders but may take on other embodiments such as pneumatic cylinders and/or linear actuators. In the alternative, rather than using cylinders, the at least one actuator may take on the form of a motor attached to a body of the of the remixing transfer vehicle 2000 and a rack (a long bar having teeth configured to interface with a pinion gear attached to a shaft

of the motor) attached to the storage cleanout door **2300**. In this latter embodiment, energizing the motor would cause its shaft and pinion gear to spin causing the storage cleanout door **2300** to move in and out by virtue of engagement of the teeth of the pinion interfacing with the teeth of the rack. Still yet, the at least one actuator may take on another form such as, but not limited to, a chain drive. As such, there are many ways one skilled in the art can configure the at least one actuator such that when the at least one actuator is energized the storage cleanout door **2300** may move outwards or inwards from a body of the remixing transfer vehicle **2000**.

[0030] In example embodiments, the storage cleanout door **2300** may further be connected to at least one guide. In example embodiments the at least one guide is embodied as a front telescoping guide **2600** and a rear telescoping guide **2650** which may each resemble a cylinder in a cylinder. The front telescoping guide **2600** may, for example, be comprised of a first cylinder **2610** which slides within a second cylinder **2620** and the rear telescoping guide **2600** may be comprised of a third cylinder **2660** which slides into a fourth cylinder **2670**. The guides **2600** and **2650** may support and guide the storage cleanout door **2300** as it is being moved under the influence of the cylinders **2500** and **2550**. Further, the guides **2600** and **2650** may aid in restraining motion of the storage cleanout door **2300** to a linear motion causing the storage cleanout door **2300** to translate in space as it moves out of its home position. In at least one nonlimiting example embodiment, the storage cleanout door **2300** moves horizontally as it translates through space under the influence of the at least one actuator. Although the exemplary front and rear telescoping guides **2600** and **2650** are shown as being comprised of a cylinder in a cylinder, this is just for exemplary purposes only. For example, the front telescoping guide **2600** and the rear telescoping guide **2650** may be comprised of tube steel of one size sliding within tube steel of another size. In another embodiment, the first cylinder **2610** may be larger than the second cylinder **2620** such that the second cylinder **2620** slides within the first cylinder **2610**. Similarly, the third cylinder **2660** may be larger than the fourth cylinder **2670** such that the fourth cylinder **2670** slides with the third cylinder **2660**. In other words, there are many ways the telescoping guides **2600** and **2650** may be configured and many ways the at least one guide may be configured.

[0031] The remixing transfer vehicle **2000** may further include front and rear door mount bars **2700** and **2750**. Additionally, the remixing transfer vehicle **2000** may include a storage frame **2800** into which an end of the storage cleanout door **2300** may insert. As shown in the various figures, the storage frame **2800** may include one or more key stocks **2810** which may guide an end of the storage cleanout door **2300** as it moves inwards and inserts into the storage frame **2800**. As in the conventional remixing transfer vehicle, the storage cleanout door **2300** may store asphalt remixed by an auger housed in the door **2300**.

[0032] FIGS. 7-12 illustrate the remixing transfer vehicle **2000** when the front and rear cylinders **2500** and **2550** are extended such that the storage cleanout door **2300** is moved out of its home position as shown in FIGS. 5 and 6. Extending the front and rear cylinders **2500** and **2550** causes the storage cleanout door **2300** to slide outwards exposing the inside of the storage cleanout door **2300** as well as the hopper without the need for a person to go underneath the second hopper of the remixing transfer vehicle **2000**. This

presents the storage cleanout door **2300** to an operator in a manner that is easy to clean (as the door may be cleaned from the side of the remixing transfer vehicle **2000** rather than underneath it). Further yet, the door is self-cleaning as it moves outwards. Additionally, the inventive design eliminates conventional supporting structures thus reducing trapping of asphalt in the remixing transfer vehicle **2000**. Further yet, accessibility to the augers is improved as is the ease of removing augers from the remixing transfer vehicle **2000**. For example, the door may be used to assist in removing the foot shaft if complete removal is necessary. Additionally, the chances of asphalt leaking out of the remixing transfer vehicle **2000** is reduced compared to the conventional remixing transfer vehicle.

[0033] Example embodiments of the invention have been described in an illustrative manner. It is to be understood that the terminology that has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of example embodiments are possible in light of the above teachings. Therefore, within the scope of the appended claims, the present invention may be practiced otherwise than as specifically described.

1. A material transfer vehicle comprising:

a first hopper arranged at a front of the material transfer vehicle, the first hopper being configured to receive asphalt from a dump truck; and

a second hopper arranged near a middle of the material transfer vehicle, the second hopper being configured to receive material from the first hopper via a first feeder, wherein the second hopper includes a storage cleanout door configured to move laterally with respect to a direction of travel of the material transfer vehicle.

2. The material transfer vehicle of claim 1, further comprising:

at least one actuator configured to move the storage cleanout door laterally with respect to the direction of travel of the material transfer vehicle.

3. The material transfer vehicle of claim 2, wherein the at least one actuator includes a front cylinder arranged near a front of the storage cleanout door and a rear cylinder arranged near a back of the storage cleanout door.

4. The material transfer vehicle of claim 1, further comprising at least one telescoping guide configured to support and guide the storage cleanout door.

5. The material transfer vehicle of claim 4, wherein the at least one telescoping guide includes a cylinder in a cylinder.

6. The material transfer vehicle of claim 4, wherein the at least one telescoping guide includes a front telescoping guide arranged near a front of the storage cleanout door and a back telescoping guide arranged near a back of the storage cleanout door.

7. The material transfer vehicle of claim 1, wherein the storage cleanout door stores asphalt remixed by an auger housed in the second hopper.

8. The material transfer vehicle of claim 1, further comprising:

a storage frame configured to receive and support an end of the storage cleanout door while the storage cleanout door is moved laterally inwards to a home position.

9. A material transfer vehicle comprising:

a hopper arranged near a middle of the material transfer vehicle;

an auger in the hopper, the auger being configured to mix asphalt in the hopper;

a bearing guard configured to support the auger;

a storage cleanout door arranged at a bottom of the hopper, the storage cleanout door being configured to translate outwards to expose a surface of the storage cleanout door;

at least one actuator configured to translate the storage cleanout door; and

at least one telescoping guide configured to support and guide the storage cleanout door.

10. The material transfer vehicle of claim **9**, wherein the at least one actuator includes a front cylinder arranged near a front of the storage cleanout door and a rear cylinder arranged near a back of the storage cleanout door.

11. The material transfer vehicle of claim **9**, wherein the at least one telescoping guide includes a cylinder in a cylinder.

12. The material transfer vehicle of claim **11**, wherein the at least one telescoping guide includes a front telescoping guide arranged near a front of the storage cleanout door and a back telescoping guide arranged near a back of the storage cleanout door.

13. The material transfer vehicle of claim **9**, wherein the storage cleanout door stores the asphalt mixed by the auger.

14. The material transfer vehicle of claim **9**, wherein the hopper is configured to receive the asphalt from a feeder that moves asphalt from a loading hopper of the material transfer vehicle.

15. A method of cleaning a hopper of a material transfer vehicle comprising:

energizing at least one actuator to cause a storage cleanout door of the hopper to move sideways and expose an inner surface of the storage cleanout door.

16. The method of claim **15**, wherein when the cleanout door moves sideways it translates.

17. The method of claim **15**, wherein the at least one actuator is at least one cylinder.

18. The method of claim **15**, wherein the at least one cylinder includes a front cylinder arranged near a front of the storage cleanout door and a rear cylinder arranged near a back of the storage cleanout door.

19. The method of claim **15**, wherein the storage cleanout door is configured to slide into and out of a storage frame.

20. The method of claim **15**, wherein when the storage cleanout door moves sideways at least one guide moves in a telescoping manner.

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